Cardiovascular System (CARD) · NPLEX-P1-CARD-01

CV-1 Cardiovascular Embryology

Brain dumps can be right in more than one way. Grade the ideas and the relationships, not the wording. A response is competent when it carries the essentials below.

NPLEX-P1-CARD-01-01-C01-BD1

Brain dump 1 · foundational retrieval: Draw the heart tube and name what each segment becomes

A competent response contains

· Sequence: cardiac progenitors pass through the cranial primitive streak into splanchnic lateral plate mesoderm and gather cranial to the neural plate as a horseshoe shaped cardiogenic field on days 16 to 18; endoderm signals of BMP and FGF with inhibition of Wnt induce NKX2-5; angioblastic cords canalize into paired endocardial tubes; lateral folding brings them to the ventral midline and they fuse into one tube by the end of week 3; cephalocaudal folding carries the tube into the future thorax; the heart beats at about day 22; looping runs from about day 23 to day 28.

· Segments from caudal to cranial: sinus venosus, primitive atrium, primitive ventricle, bulbus cordis, truncus arteriosus opening into the aortic sac.

· Derivatives: truncus arteriosus gives the proximal ascending aorta and pulmonary trunk; conus cordis gives the smooth outflow tracts of both ventricles; proximal bulbus cordis gives the trabeculated right ventricle; primitive ventricle gives most of the trabeculated left ventricle; primitive atrium gives the trabeculated atria including the auricles; right horn of the sinus venosus gives the smooth right atrium, the sinus venarum; left horn gives the coronary sinus and the oblique vein of the left atrium; the primitive pulmonary vein gives the smooth part of the left atrium.

· The primary heart field forms the initial tube, mainly the left ventricle and parts of the atria; the secondary heart field in pharyngeal mesoderm adds most of the right ventricle and the outflow tract, meaning the conus cordis and truncus arteriosus, plus part of the atria and the sinus venosus.

· Too few secondary field cells leave the outflow tract too short to rotate and align, which is one cause of outflow defects.

· Wall layers: inner endocardium, cardiac jelly that becomes the cushions, and outer myocardium; the proepicardium later supplies the epicardium, the coronary vessels and many cardiac fibroblasts.

· Loss of the dorsal mesocardium leaves the transverse pericardial sinus.

· Rough or trabeculated walls come from the original atrial and ventricular segments; smooth walls come from incorporated veins or from the outflow tract.

Distinctions that matter

· The two heart fields contribute different chambers, so a right ventricular or outflow problem points at the secondary field while the left ventricle comes mainly from the primary field.

· Smooth against trabeculated wall is a statement about embryonic origin, not about function.

Common misconceptions

· Writing that the whole heart arises from one field. The outflow tract and most of the right ventricle are added later by the secondary heart field.

· Treating the cardiac jelly as packing material. It is the tissue the endocardial cushions form in, so the valves and central septa come from it.

Where it is taught: Section C01, opening paragraphs and the table of segments and derivatives of the CV-1 Cardiovascular Embryology notes.

NPLEX-P1-CARD-01-01-C01-BD2

Brain dump 2 · integration and application: A child with chronic sinusitis and an unusual chest film

A competent response contains

· The defect is in motile cilia, usually in the dynein arms, and the condition is primary ciliary dyskinesia, usually inherited in an autosomal recessive pattern.

· Nonfunctional dynein stops the rotary beat of the tilted monocilia at the primitive node, so there is no leftward nodal flow; NODAL, LEFTY2 and PITX2 are still expressed but on a side chosen at random, so sidedness is unpredictable rather than reversed.

· About half of patients have situs inversus totalis with dextrocardia, most of the remainder have normal situs, and a minority have heterotaxy, also called situs ambiguus.

· Kartagener syndrome is situs inversus with chronic sinusitis and bronchiectasis.

· Immotile airway cilia cannot clear mucus, giving neonatal respiratory distress, a wet cough, otitis media and bronchiectasis; immotile sperm flagella cause male infertility, and poor tubal transport raises the risk of ectopic pregnancy.

· Risk of structural heart disease depends on whether the heart and the body agree about sidedness: mirror image dextrocardia keeps the connections concordant so the heart is usually normal, while dextrocardia with normal abdominal situs, or heterotaxy with asplenia or polysplenia, carries a high risk of complex lesions.

· Mirror image dextrocardia inverts the P wave, the QRS complex and the T wave in lead I and gives diminishing R waves across the precordial leads.

Distinctions that matter

· Randomized sidedness is not the same as reversed sidedness, which is why only about half of these patients have mirrored organs.

· Concordance between heart and viscera, not the position of the heart by itself, is what predicts complex defects.

Common misconceptions

· Saying the cilia problem causes the heart defect directly. The cilia set the left and right signal, and the defects follow from discordant sidedness.

· Expecting every patient with this ciliary defect to have situs inversus. Most of those without it have normal situs.

Where it is taught: Section C01, paragraphs on primary ciliary dyskinesia and on concordance of situs of the CV-1 Cardiovascular Embryology notes.

NPLEX-P1-CARD-01-01-C01-RT01

Retrieval exercise: Rebuild the left and right signalling sequence

A competent response contains

· Step 1: dynein drives rotation of the tilted monocilia at the primitive node.

· Step 2: that rotation moves fluid leftward across the node.

· Step 3: NODAL and LEFTY2 are induced in left lateral plate mesoderm, and midline LEFTY1 forms a barrier that stops the signal spreading to the right.

· Step 4: NODAL induces PITX2 on the left.

· Step 5: PITX2 specifies left sided identity of the heart, lungs and gut, and the heart tube loops to the right.

· The cilia are tilted monocilia on the cells of the primitive node and the motor protein is dynein.

· If nodal flow fails, steps 3 to 5 still occur but on a randomly chosen side.

· In the loop the bulboventricular part bends ventrally, caudally and to the right while the atrial part moves dorsally, cranially and to the left, so the atria end up above and behind the ventricles, with the primitive ventricle, the future left ventricle, on the left and the bulbus cordis, the future right ventricle, on the right.

Distinctions that matter

· The midline LEFTY1 barrier is what keeps the pathway one sided; without it the left signal would cross over.

· Looping is a change in the position of segments already present, not the addition of new chambers.

Common misconceptions

· Writing that loss of nodal flow switches the signal to the right side. It removes the instruction, so the side is chosen at random.

· Placing the future left ventricle on the right after looping. The primitive ventricle ends up on the left.

Where it is taught: Section C01, flow diagram on left and right signalling and the looping paragraph of the CV-1 Cardiovascular Embryology notes.

NPLEX-P1-CARD-01-01-C02-BD1

Brain dump 1 · foundational retrieval: Draw atrial septation in four panels

A competent response contains

· Septum primum grows from the atrial roof toward the endocardial cushions, leaving the ostium primum as a gap below its free edge.

· Apoptosis in the upper part of the septum primum opens the ostium secundum, a hole inside that septum.

· The septum primum then fuses with the cushions and the ostium primum closes at the end of week 5.

· The thick septum secundum grows down on the right side of the septum primum, and its free edge bounds the foramen ovale, which is the gap left by that second septum rather than a hole within it.

· The thin septum primum lies on the left and the rigid septum secundum on the right, and their openings are not aligned, so the lower part of the septum primum works as a flap valve.

· Before birth, higher right atrial pressure pushes the flap open and blood crosses right to left only; after birth higher left atrial pressure presses the flap shut.

· The adult remnant is the fossa ovalis: its floor is septum primum and its rim, the limbus, is the edge of the septum secundum.

· Anatomic fusion takes months.

Distinctions that matter

· The ostium secundum is a hole in the septum primum, while the foramen ovale is the gap left by the septum secundum; the names are easy to swap and the structures are not the same.

· Thin and left is septum primum, thick and right is septum secundum, and the offset is what makes the valve one way.

Common misconceptions

· Describing the foramen ovale as a hole in a single septum. It is an opening between two overlapping septa.

· Saying the flap closes because the hole shrinks. It closes because the pressure gradient reverses and presses the thin septum against the thick one.

Where it is taught: Section C02, flow diagram on atrial septation and the following paragraphs of the CV-1 Cardiovascular Embryology notes.

NPLEX-P1-CARD-01-01-C02-BD2

Brain dump 2 · integration and application: An unsealed flap against a true hole

A competent response contains

· Patent foramen ovale: no tissue is missing, the septa simply never fused, and this persists in about a quarter of adults; blood crosses only when right atrial pressure transiently exceeds left, as on release of a Valsalva strain, with coughing, or with acute pulmonary hypertension from pulmonary embolism, and the consequence is paradoxical embolism with stroke.

· Secundum atrial septal defect: a true tissue deficiency at the fossa ovalis from excess resorption of the septum primum or a deficient septum secundum; because the right ventricle is more compliant the shunt runs left to right.

· The extra volume loads the right atrium, right ventricle and pulmonary circulation; the left ventricle is spared because the shunt leaves before it.

· The murmur is a systolic ejection murmur at the left upper sternal border, produced by increased flow across a normal pulmonary valve rather than by flow through the defect; the interatrial pressure difference is small, so flow across the defect itself is nearly silent.

· A mid diastolic rumble across the tricuspid valve indicates a large shunt.

· The tracing shows right axis deviation with an rsR prime pattern in V1, which follows from right ventricular volume loading.

· Reversal sequence: high pulmonary flow over decades remodels the pulmonary arterioles, pulmonary vascular resistance rises, pulmonary pressure exceeds left sided pressure, the shunt reverses to right to left, and cyanosis appears; this is Eisenmenger syndrome and can end any large left to right shunt.

· The second sound is widely split because the right ventricle takes systemic return plus the shunt on every beat, so pulmonic closure is always late; it is fixed because inspiration raises right atrial pressure and shrinks the shunt by about the amount the systemic return rises, the atria behaving as one chamber, so right ventricular filling and the timing of pulmonic closure barely change with respiration.

· Right bundle branch block delays pulmonic closure electrically, so the wide split still varies with breathing.

· Known genetic causes of secundum defects are TBX5 mutation in Holt-Oram syndrome with upper limb defects and NKX2-5 mutation with atrioventricular conduction disease.

Distinctions that matter

· An unsealed flap needs a transient pressure reversal before anything crosses; a true deficiency shunts on every beat.

· Fixed splitting comes from a shunt that cancels the respiratory change, while a wide but moving split comes from delayed electrical activation of the right ventricle.

Common misconceptions

· Attributing the murmur to flow through the atrial defect. The gradient there is too small, and the sound comes from the extra flow across the pulmonary valve.

· Expecting left ventricular volume overload in an atrial septal defect. The shunt leaves the circuit before the left ventricle, so the right heart and lungs carry the load.

Where it is taught: Section C02, paragraphs on the patent foramen ovale and the secundum defect, and the worked example on the second heart sound of the CV-1 Cardiovascular Embryology notes.

NPLEX-P1-CARD-01-01-C03-BD1

Brain dump 1 · foundational retrieval: What the endocardial cushions build

A competent response contains

· A single atrioventricular canal opens mainly over the future left ventricle and must shift to the right before dividing.

· At the end of week 4 the atrioventricular canal and outflow myocardium signal the endocardium through the TGF beta and BMP families; endocardial cells lose their junctions and invade the cardiac jelly, a change called epithelial to mesenchymal transformation, becoming the cushion mesenchyme.

· Superior and inferior atrioventricular cushions form with smaller lateral cushions; the outflow ridges form separately and receive neural crest cells.

· In week 5 the superior and inferior cushions fuse and divide the canal into right and left orifices.

· The fused mass closes the ostium primum, the lowest part of the atrial septum, at the end of week 5.

· Inferior cushion tissue joins the conal ridges and the muscular septum to close the interventricular foramen as the membranous septum by about the end of week 7.

· Cushion mesenchyme around each orifice is undermined on its ventricular side; the muscular strands left behind become the chordae tendineae, with muscle replaced by dense connective tissue, and the papillary muscles. The left orifice ends with two leaflets and the right with three.

· The muscular ventricular septum forms as the expanding ventricles press their medial walls together.

· The membranous septum closes last and needs three tissue sources to meet, the conal ridges, the muscular septum and the inferior cushion, which is why most ventricular septal defects are perimembranous; muscular defects, sometimes multiple, are less common and small ones often close.

Distinctions that matter

· The cushions build the valves, the lowest atrial septum and part of the ventricular septum, so they are central structures rather than valve tissue alone.

· The muscular septum forms by chamber growth, while the membranous part forms by three tissues meeting, and that difference explains where defects sit.

Common misconceptions

· Saying the valve leaflets grow out of the ventricular wall. They are carved out of cushion tissue by undermining from the ventricular side.

· Treating the chordae as tendon from the start. They are muscular strands in which the muscle is replaced by dense connective tissue.

Where it is taught: Section C03, first three paragraphs of the CV-1 Cardiovascular Embryology notes.

NPLEX-P1-CARD-01-01-C03-BD2

Brain dump 2 · integration and application: Why one cushion failure damages several structures

A competent response contains

· The failed step is fusion of the superior and inferior atrioventricular cushions, giving an atrioventricular septal defect, also called an atrioventricular canal defect.

· Because the cushions build several structures, one failure leaves a primum atrial septal defect, an inlet ventricular septal defect and a single common atrioventricular valve with five leaflets in the complete form.

· Failure at three months follows from pressure as well as flow: the ventricular defect transmits left ventricular pressure to the pulmonary arteries, so the lungs see high flow at high pressure, and pulmonary vascular disease arrives earlier than with an isolated atrial defect, where the shunt is at low atrial pressure.

· The atrioventricular node and His bundle are displaced inferiorly and posteriorly and the left anterior fascicle is short, giving left or superior axis deviation, often with first degree atrioventricular block.

· A secundum defect at the fossa ovalis gives right axis deviation instead, because the right ventricle is volume loaded and the conduction tissue sits normally.

· Partial form: two separate orifices with a primum atrial septal defect and a cleft anterior mitral leaflet producing mitral regurgitation.

· About 40 to 50 percent of children with trisomy 21 have congenital heart disease, and the complete atrioventricular septal defect is the characteristic lesion; the extra chromosome is thought to disturb cushion cell adhesion, migration and matrix.

· Ebstein anomaly: the septal and posterior tricuspid leaflets are poorly undermined, so their attachment is displaced toward the apex, part of the right ventricle is atrialized and the valve leaks.

· Tricuspid atresia: the right atrioventricular orifice is obliterated, so right atrial blood must cross the atrial septum, the hypoplastic right ventricle fills through a ventricular septal defect, and the lungs are supplied through that defect or through the ductus.

Distinctions that matter

· Left or superior axis deviation points at cushion failure with displaced conduction tissue; right axis deviation points at a volume loaded right ventricle with a normally placed node.

· A defect that carries ventricular pressure into the lungs behaves differently from one that carries only atrial pressure, even at the same shunt volume.

Common misconceptions

· Grouping all holes between the atria together. Only cushion defects involve the atrioventricular valves and give left axis deviation.

· Expecting a cushion defect to present like an atrial defect. The added ventricular communication brings heart failure and pulmonary vascular disease in months.

Where it is taught: Section C03, paragraph on atrioventricular septal defect, the comparison table, the trisomy 21 paragraph and the final paragraph of the CV-1 Cardiovascular Embryology notes.

NPLEX-P1-CARD-01-01-C03-CP01

Comparison table: Three openings between the atria compared

A competent response contains

· Developmental step that failed: patent foramen ovale, the two septa do not fuse after birth; secundum defect, excess resorption of the septum primum or a deficient septum secundum; complete atrioventricular septal defect, the atrioventricular cushions fail to fuse.

· Tissue missing: no tissue loss in the patent foramen ovale; a true deficiency in the secundum defect; tissue missing in the lower septum and the inlet ventricular septum in the cushion defect.

· Location: patent foramen ovale between the overlapping septa; secundum defect at the fossa ovalis; cushion defect low in the septum above the atrioventricular valves, with a primum atrial defect and an inlet ventricular defect.

· Atrioventricular valves: normal in the patent foramen ovale and the secundum defect; a single common five leaflet valve in the complete cushion defect, and a cleft anterior mitral leaflet with regurgitation in the partial form.

· Flow: patent foramen ovale crosses right to left only when right atrial pressure transiently exceeds left; secundum defect shunts left to right every beat because the right ventricle is more compliant; complete cushion defect shunts left to right at both atrial and ventricular level, carrying left ventricular pressure to the lungs.

· Axis: normal in the patent foramen ovale; right axis deviation with rsR prime in V1 in the secundum defect; left or superior axis deviation, often with first degree block, in the cushion defect.

· Timing: the patent foramen ovale is silent until a transient pressure reversal allows paradoxical embolism in adult life; the secundum defect is often silent into adulthood and can end in Eisenmenger syndrome after decades; the complete cushion defect causes heart failure within months with early pulmonary vascular disease.

· Associations: Holt-Oram syndrome with TBX5 and conduction disease with NKX2-5 in the secundum defect; trisomy 21 in the complete cushion defect; none specific for the patent foramen ovale.

Distinctions that matter

· Only the patent foramen ovale lacks tissue loss, and only the cushion defects involve the atrioventricular valves and give left axis deviation.

· Direction of flow in the patent foramen ovale is right to left and intermittent, while the two true defects shunt left to right continuously until resistance changes.

Common misconceptions

· Calling a patent foramen ovale a small atrial septal defect. No septal tissue is missing and the physiology is different.

· Using murmur loudness to judge shunt size at the atrial level. The gradient is small, and the diastolic tricuspid rumble is what signals a large shunt.

Where it is taught: Section C03, table of interatrial and atrioventricular septal communications, with Section C02 for the patent foramen ovale and the secundum defect of the CV-1 Cardiovascular Embryology notes.

NPLEX-P1-CARD-01-01-C04-BD1

Brain dump 1 · foundational retrieval: Draw the spiral outflow septum and the three defects

A competent response contains

· The outflow tract must become a pulmonary trunk leaving the right ventricle anteriorly and to the left and an aorta leaving the left ventricle posteriorly and to the right.

· In week 5 the right superior and left inferior truncal ridges grow distally while swinging around the truncal wall, so the aorticopulmonary septum they form twists about 180 degrees.

· The right dorsal and left ventral conal ridges fuse into the conal or outlet septum, which separates the right ventricular infundibulum from the aortic vestibule and meets the muscular septum and the inferior cushion to complete the membranous septum.

· Cardiac neural crest cells come from the hindbrain and migrate through pharyngeal arches 3, 4 and 6; some support the arch arteries and help form the thymus and parathyroids, and others populate the truncal and conal ridges and the semilunar valves. Removing this crest in animal embryos produces persistent truncus arteriosus.

· Ridges that never form or fuse give persistent truncus arteriosus: one trunk and one valve sitting over a ventricular septal defect, supplying the coronary, pulmonary and systemic arteries, with mixing and mild cyanosis and then excess pulmonary flow and heart failure as resistance falls.

· A septum that fails to spiral gives D transposition of the great arteries: the aorta leaves the right ventricle and the pulmonary trunk leaves the left ventricle, the circuits run in parallel, survival requires mixing through the foramen ovale, the ductus or a ventricular septal defect, cyanosis is severe and early, the vascular pedicle is narrow, and maternal diabetes is a risk factor.

· A septum displaced anterosuperiorly gives tetralogy of Fallot: pulmonary stenosis, a malalignment ventricular septal defect, an overriding aorta and right ventricular hypertrophy.

· Because the outlet septum and the ventricular septum must align, displacing the outlet septum leaves a malalignment defect in the ventricular septum.

Distinctions that matter

· No septum, a straight septum and a displaced septum produce the three classic conotruncal defects, so naming the step tells you the anatomy.

· Truncal ridges divide the great vessels while conal ridges divide the outflow tracts of the ventricles, and they fail with different consequences.

Common misconceptions

· Treating transposition as a problem of vessel position alone. The septum formed but did not spiral, which is why each vessel sits over the wrong ventricle.

· Listing the four features of tetralogy as four separate faults. One displaced septum produces all four.

Where it is taught: Section C04, first two paragraphs and the table of conotruncal defects of the CV-1 Cardiovascular Embryology notes.

NPLEX-P1-CARD-01-01-C04-BD2

Brain dump 2 · integration and application: Two blue newborns from two different failures

A competent response contains

· First newborn: D transposition of the great arteries, from a septum that failed to spiral, so the aorta arises from the right ventricle and the pulmonary trunk from the left ventricle.

· The circuits run in parallel, so systemic venous blood returns to the body and pulmonary venous blood returns to the lungs, and the two can only supply each other through a shunt: the foramen ovale, the ductus arteriosus or a ventricular septal defect.

· What changed on day two is that the ductus arteriosus closed as arterial oxygen rose and prostaglandin E2 fell, so mixing depended on the interatrial opening alone and systemic oxygen delivery fell.

· The great vessels lie one in front of the other, which narrows the vascular pedicle on the film.

· Second child: tetralogy of Fallot, from anterosuperior displacement of the infundibular septum, giving pulmonary stenosis, a malalignment ventricular septal defect, an overriding aorta and right ventricular hypertrophy; the raised apex with a concave upper left border is the boot shaped heart.

· Because the ventricular defect is large, both ventricles share systolic pressure, so shunt direction depends on right ventricular outflow resistance against systemic resistance; mild obstruction gives a mostly left to right shunt and a pink child.

· The murmur arises from the narrowed right ventricular outflow tract, so it softens during a spell because less blood is crossing that outflow.

· Chronic hypoxaemia causes secondary polycythaemia and clubbing.

· The 22q11.2 deletion, called DiGeorge or velocardiofacial syndrome, removes one copy of TBX1 and neighbouring genes and disrupts the pharyngeal tissue through which crest cells migrate, so outflow defects such as truncus arteriosus, tetralogy and interrupted aortic arch arrive with thymic hypoplasia and T cell deficiency, parathyroid hypoplasia with hypocalcaemia, and facial and palatal anomalies.

· Excess retinoic acid, as with maternal isotretinoin, disturbs crest migration and causes conotruncal and craniofacial defects.

Distinctions that matter

· Cyanosis in transposition is fixed by the anatomy of parallel circuits, while cyanosis in tetralogy varies with the balance of two resistances.

· A quieter murmur means less flow across the obstruction, which in tetralogy means a worse spell rather than a better heart.

Common misconceptions

· Reading a louder murmur as more severe disease in tetralogy. The murmur fades as the spell deepens.

· Expecting every child with tetralogy to be blue from birth. With mild outflow obstruction the shunt runs mostly left to right.

Where it is taught: Section C04, table of conotruncal defects, the paragraphs on transposition and tetralogy, and the paragraph on the 22q11.2 deletion of the CV-1 Cardiovascular Embryology notes.

NPLEX-P1-CARD-01-01-C04-PR01

Prediction exercise: Predict what happens during a spell and during squatting

A competent response contains

· Systemic vascular resistance falls while the child cries or exerts.

· Right ventricular outflow resistance therefore rises relative to systemic resistance, because the outflow obstruction is fixed while the systemic bed dilates.

· Blood crosses the ventricular septal defect right to left, and more of it crosses, because the large defect lets both ventricles share pressure and blood takes the lower resistance route.

· Pulmonary blood flow falls and arterial oxygen saturation falls with it.

· The murmur softens, because it is generated by flow through the narrowed right ventricular outflow tract and less blood is crossing it.

· Hypoxaemia and acidosis lower systemic resistance further, which drives more right to left shunting, so the spell feeds itself.

· Squatting kinks the femoral arteries and raises systemic resistance, so the shunt swings back toward left to right, pulmonary flow and saturation rise, the murmur becomes louder again and the cycle is broken.

Distinctions that matter

· Shunt direction follows the comparison between two resistances, not the size of the defect, because the defect is large enough to equalize pressure.

· Murmur loudness tracks flow across the obstruction, so it moves in the opposite direction to the severity of the spell.

Common misconceptions

· Expecting the obstruction itself to worsen during a spell. The fixed obstruction stays the same while the systemic resistance falls around it.

· Reading squatting as an effect on the lungs. It acts on the systemic circulation by raising its resistance.

Where it is taught: Section C04, cycle diagram on the hypercyanotic spell and the preceding paragraph on shunt direction of the CV-1 Cardiovascular Embryology notes.

NPLEX-P1-CARD-01-01-C05-BD1

Brain dump 1 · foundational retrieval: Draw the fetal circulation with its saturations

A competent response contains

· Route: umbilical vein to the liver, part of the flow through the ductus venosus into the inferior vena cava, right atrium, across the foramen ovale to the left atrium, left ventricle, ascending aorta with the coronary, carotid and subclavian branches first; superior vena cava blood to the right atrium, right ventricle and pulmonary trunk, most of it across the ductus arteriosus into the descending aorta, then back through the umbilical arteries to the placenta.

· The umbilical vein is the most saturated fetal vessel at about 80 percent, with a partial pressure of oxygen of only about 30 to 35 mm Hg; the umbilical arteries carry blood at about 60 percent saturation.

· The crista dividens, the lower edge of the septum secundum, with the valve of the inferior vena cava, streams the better oxygenated inferior caval blood across the foramen ovale.

· Pulmonary vascular resistance is high because the lungs are fluid filled, their vessels are compressed, low alveolar oxygen tension causes hypoxic vasoconstriction, and mediators favour constriction with high endothelin and low nitric oxide.

· Systemic resistance is low because the placenta is a large low resistance circuit.

· Pulmonary trunk pressure therefore exceeds descending aortic pressure, which is why most right ventricular output crosses the ductus arteriosus and only a small fraction of combined output passes through the lungs.

· The right ventricle supplies slightly more than half of the combined output, which is why both ventricles have similar wall thickness at birth.

· Three shunts: ductus venosus past the liver, foramen ovale between the atria, ductus arteriosus from pulmonary trunk to descending aorta.

Distinctions that matter

· Saturation and partial pressure are separate numbers here: the umbilical vein is about 80 percent saturated at a partial pressure an adult would not tolerate, because fetal haemoglobin binds oxygen more tightly.

· Both ventricles pump into the systemic circuit in the fetus, which is what parallel circulation means.

Common misconceptions

· Describing the umbilical vein as fully saturated. It is the best fetal blood at about 80 percent, not arterial adult blood.

· Expecting a thin right ventricle before birth. It carries slightly more than half of the combined output, so the walls are of similar thickness.

Where it is taught: Section C05, opening paragraphs, flow diagram on the fetal circulation and the following paragraphs of the CV-1 Cardiovascular Embryology notes.

NPLEX-P1-CARD-01-01-C05-BD2

Brain dump 2 · integration and application: Why the fetal brain is well supplied at a low oxygen tension

A competent response contains

· Four reasons oxygenation works: fetal haemoglobin has a higher oxygen affinity, haemoglobin concentration is higher, cardiac output is high for body mass, and streaming delivers the best blood to the heart and brain.

· Fetal haemoglobin has two alpha and two gamma chains; the gamma chain carries serine where the beta chain has the 2,3 bisphosphoglycerate binding histidine at position 143, so 2,3 bisphosphoglycerate binds weakly and the half saturation pressure is about 19 mm Hg against about 27 mm Hg for adult haemoglobin.

· The double Bohr effect at the placenta: fetal blood loses carbon dioxide and gains affinity while maternal blood gains carbon dioxide and loses affinity, so oxygen moves from mother to fetus.

· Best oxygenated route: umbilical vein, ductus venosus, inferior vena cava, streamed across the foramen ovale by the crista dividens, left atrium with only slight dilution by the small pulmonary venous return, left ventricle, ascending aorta, then the coronary arteries and the carotid and subclavian arteries before the ductus joins the aorta distal to the left subclavian artery.

· The lower body receives the mixture of left ventricular blood and the desaturated ductal blood from the right ventricle, which carries superior caval and coronary sinus blood with part of the inferior caval stream.

· If the ductus arteriosus narrows in mid gestation the right ventricle has to eject into the high resistance pulmonary bed, so right ventricular pressure load rises, while flow reaching the descending aorta falls, so the lower body depends more on left ventricular output.

· If the foramen ovale narrows instead, the left atrium and left ventricle lose most of their filling and grow poorly while the right heart enlarges.

· Embryonic chambers grow in proportion to the flow that passes through them.

Distinctions that matter

· The shift in the dissociation curve is a property of the gamma chain and its weak 2,3 bisphosphoglycerate binding, not of a different oxygen binding site.

· Streaming is an anatomical effect inside the right atrium, separate from the affinity effect in the blood.

Common misconceptions

· Saying the fetal brain receives placental blood directly. It receives left ventricular blood that was streamed across the foramen ovale and only slightly diluted.

· Expecting a narrowed atrial opening to enlarge the left heart because pressure rises. Filling falls, so the chamber grows poorly.

Where it is taught: Section C05, paragraphs on streaming, on the lower body supply and on fetal haemoglobin, with the stop and predict box in Section C02 of the CV-1 Cardiovascular Embryology notes.

NPLEX-P1-CARD-01-01-C05-RT02

Retrieval exercise: Rebuild the path of blood through the fetus

A competent response contains

· Step 1: the umbilical vein brings the best oxygenated blood, about 80 percent saturated, from the placenta.

· Step 2: the ductus venosus shunts part of that flow past the hepatic sinusoids into the inferior vena cava; the rest enters the sinusoids through the left portal vein.

· Step 3: the crista dividens streams inferior caval blood through the foramen ovale into the left atrium, which is possible because right atrial pressure exceeds left atrial pressure in the fetus.

· Step 4: the left ventricle ejects into the ascending aorta, and the coronary, carotid and subclavian arteries branch before any mixing point.

· Step 5: superior caval blood passes right atrium to right ventricle to pulmonary trunk.

· Step 6: most pulmonary trunk flow crosses the ductus arteriosus to the descending aorta, because pulmonary vascular resistance is high and pulmonary trunk pressure exceeds descending aortic pressure.

· Step 7: the umbilical arteries, branches of the internal iliac arteries, return blood at about 60 percent saturation to the placenta.

· The two shunt steps are step 2, bypassing the liver, and step 6, bypassing the lungs; the foramen ovale in step 3 bypasses the right ventricle and the lungs as well.

· The inlet of the ductus venosus regulates how much umbilical flow bypasses the liver and protects the heart from sudden volume loads such as those during uterine contractions.

· Inferior caval blood goes to the left heart and the brain; superior caval blood goes to the right heart, the ductus and the lower body.

Distinctions that matter

· Each step follows a pressure or resistance gradient, so the path is not a set of leaks but the route of least resistance at that moment.

· The ductus venosus bypasses the liver while the ductus arteriosus bypasses the lungs, and the similar names are easy to swap.

Common misconceptions

· Writing that all umbilical venous blood bypasses the liver. Part of it perfuses the hepatic sinusoids through the left portal vein.

· Treating the foramen ovale as a passive hole. Streaming by the crista dividens is what sends the better oxygenated blood through it.

Where it is taught: Section C05, flow diagram on the path of blood through the fetal circulation and the two paragraphs that follow it of the CV-1 Cardiovascular Embryology notes.

NPLEX-P1-CARD-01-01-C06-BD1

Brain dump 1 · foundational retrieval: Rebuild the changes at birth and the remnants table

A competent response contains

· Step 1: the lungs expand with the first breaths and alveolar oxygen tension rises.

· Step 2: pulmonary vascular resistance falls and pulmonary flow rises several fold, because hypoxic vasoconstriction is relieved and because oxygen and shear stress release nitric oxide and prostacyclin.

· Step 3: the cord is clamped, systemic resistance rises and inferior caval return falls.

· Step 4: left atrial pressure exceeds right atrial pressure, so the septum primum is pressed against the septum secundum and the foramen ovale closes functionally within minutes to hours; anatomic fusion takes months.

· Step 5: rising arterial oxygen tension and falling prostaglandin E2 constrict the ductus arteriosus.

· Step 6: the ductus venosus and the umbilical vessels close and fibrose into ligaments.

· Ductal oxygen mechanism: oxygen sensitive potassium channels close, the membrane depolarizes and L type calcium channels open, so the ductal smooth muscle contracts.

· Prostaglandin mechanism: prostaglandin E2 falls because the placenta is gone and the perfused lungs metabolize it; in the fetus prostaglandin E2 acting on EP receptors raised cyclic AMP and kept the muscle relaxed. Functional ductal closure takes 1 to 3 days in term infants and fibrosis over weeks leaves the ligamentum arteriosum.

· Across the ductus the gradient reverses first, so flow briefly runs aorta to pulmonary trunk; the umbilical arteries constrict within minutes to mechanical, thermal and oxygen stimuli, and the umbilical vein and ductus venosus close slightly later, after which the liver is perfused by the portal vein and hepatic artery.

· Remnants: umbilical vein, placenta to liver, becomes the ligamentum teres hepatis in the free edge of the falciform ligament; ductus venosus, umbilical vein to inferior vena cava past the sinusoids with a regulated inlet, becomes the ligamentum venosum in the fissure between the caudate and left lobes; foramen ovale, right to left atrium by streaming and a right atrial pressure above left, becomes the fossa ovalis on the right side of the interatrial septum; ductus arteriosus, pulmonary trunk to descending aorta driven by high pulmonary resistance, low oxygen tension and prostaglandin E2, becomes the ligamentum arteriosum from

the left pulmonary artery to the underside of the aortic arch; umbilical arteries, internal iliac arteries to placenta, become the medial umbilical ligaments distally and the superior vesical arteries proximally; the allantois and urachus, never a vessel, become the median umbilical ligament from the bladder apex to the umbilicus.

· Pressure closes the foramen ovale, while chemical signals of oxygen and prostaglandin E2 close the ductus arteriosus.

Distinctions that matter

· The foramen ovale closes mechanically and the ductus arteriosus chemically, which is why one shuts in minutes and the other over days.

· The ligamentum venosum and the ligamentum arteriosum come from different shunts, and the paired medial umbilical ligaments are not the single median umbilical ligament.

Common misconceptions

· Saying ductal flow stops the moment the cord is clamped. The gradient reverses first and flow briefly runs aorta to pulmonary trunk before the muscle constricts.

· Treating functional closure as permanent. Anatomic closure takes weeks for the ductus and months for the foramen ovale.

Where it is taught: Section C06, opening paragraphs, flow diagram on closure of the shunts and the table of fetal shunts and remnants of the CV-1 Cardiovascular Embryology notes.

NPLEX-P1-CARD-01-01-C06-BD2

Brain dump 2 · integration and application: A vessel that stays open and the lesions that depend on it

A competent response contains

· A patent ductus arteriosus is most common in prematurity because immature ductal muscle responds less to oxygen and more to prostaglandin E2; congenital rubella and high altitude also predispose.

· Aortic pressure exceeds pulmonary pressure throughout the cardiac cycle, so flow never stops and the murmur is continuous, a machinery murmur heard below the left clavicle.

· Diastolic runoff into the pulmonary artery widens the pulse pressure, and the shunt volume loads the left heart.

· Ductus dependent lesions: transposition of the great arteries depends on the ductus for mixing, pulmonary atresia depends on it for lung blood flow, and hypoplastic left heart or critical coarctation depends on it for body blood flow.

· Differential cyanosis in a long standing large ductus: years of high flow and pressure remodel the pulmonary arterioles until pulmonary pressure exceeds aortic pressure and the shunt reverses, which is Eisenmenger physiology; because the ductus enters the aorta distal to the left subclavian artery, the head and arms still receive unmixed left ventricular blood while the descending aorta carries desaturated blood to the lower body, so the toes are cyanotic and clubbed while the fingers stay pink.

· A reversed shunt at ventricular or atrial level mixes blood before the aorta, so fingers and toes are equally cyanotic.

· In persistent pulmonary hypertension of the newborn the pulmonary resistance stays high despite normal anatomy, so blood shunts right to left through the foramen ovale and the ductus arteriosus.

Distinctions that matter

· A continuous murmur reports a gradient that lasts all cycle, which is what separates ductal flow from a systolic shunt between the ventricles.

· Where the shunt sits relative to the arch branches decides whether cyanosis is differential or uniform.

Common misconceptions

· Expecting a ductus to load the right heart. The shunt leaves the aorta and returns through the lungs to the left atrium, so the left heart carries the volume.

· Reading cyanosis with normal heart anatomy as a structural defect. Persistent pulmonary hypertension of the newborn shunts right to left through normal fetal channels.

Where it is taught: Section C06, paragraph on patent ductus arteriosus and the worked example on differential cyanosis of the CV-1 Cardiovascular Embryology notes.

NPLEX-P1-CARD-01-01-C06-CA01

Clinical application: A newborn who collapses on the third day

A competent response contains

· Recognize: the ductus arteriosus has closed. Until the third day it was carrying right ventricular blood into the descending aorta and supplying the lower body past an obstruction in the aorta, which is the pattern of a critical coarctation or an otherwise ductus dependent systemic circulation.

· Explain: in the fetus and in the first days the lower body is supplied through the ductus, so femoral pulses and perfusion were normal on day one. As the ductus constricted over 1 to 3 days the only route to the lower body became the obstructed aorta, so the femoral pulses disappeared, systemic perfusion fell with pallor and mottling, the left ventricle faced a sudden pressure load, left atrial pressure and pulmonary venous pressure rose to give fast breathing and congested lung fields, and systemic venous congestion enlarged the liver.

· Differentiate: a newborn with parallel circuits in transposition is deeply cyanotic within hours with poor saturation everywhere and a narrow vascular pedicle, not a pressure difference between upper and lower body; a newborn whose pulmonary resistance never fell is cyanotic with a structurally normal heart and shunts right to left at both the foramen ovale and the ductus, again without absent femoral pulses; this baby has preserved upper body saturation and pressure with absent femoral pulses and a saturation gradient, which points to an obstruction in the aorta distal to the arch branches.

· Investigate: compare pulses and blood pressure in the right arm with the legs, expecting a strong upper limb pulse with raised or normal pressure and absent or weak femoral pulses with a pressure gradient between arm and leg; compare pre and post ductal saturations, which here are 96 percent at the right hand and 82 percent at the right foot; palpate the liver edge, enlarged at 3 cm below the costal margin; and note the large heart with congested lung fields on the film.

· Decide: the ductus is normally closed by rising arterial oxygen tension, which shuts oxygen sensitive potassium channels and opens L type calcium channels, and by falling prostaglandin E2 once the placenta is gone and the perfused lungs metabolize it. The prostaglandin E2 supply was lost first, at delivery with the placenta, while the oxygen driven constriction then acted over the next days, and the lower body circulation now depends entirely on flow through the obstructed aorta.

· Justify: the ductus enters the aorta distal to the left subclavian artery, so the coronary, carotid and subclavian branches leave the arch before that point and carry fully saturated left ventricular blood, giving 96 percent at the right hand, while the descending aorta beyond the ductal insertion receives desaturated right ventricular blood through the ductus, giving 82 percent at the foot.

Distinctions that matter

· A saturation and pulse difference between upper and lower body localizes the problem distal to the arch branches, while uniform cyanosis points to mixing before the aorta.

· Loss of prostaglandin E2 happens at delivery, whereas the rise in oxygen tension acts over the following days, which is why collapse comes on day two or three rather than at birth.

Common misconceptions

· Reading a normal day one examination as evidence against a structural lesion. The ductus was supplying the lower body and hiding the obstruction.

· Expecting cyanosis to be the first sign in a ductus dependent systemic circulation. Pallor, absent femoral pulses and failure come first, with only a modest saturation gradient.

Where it is taught: Section C06, paragraphs on circulatory changes at birth, ductal closure and ductus dependent lesions, with the worked example on differential cyanosis of the CV-1 Cardiovascular Embryology notes.

NPLEX-P1-CARD-01-01-C07-BD1

Brain dump 1 · foundational retrieval: Build the arch artery table and draw both nerve loops

A competent response contains

· Each pharyngeal arch artery runs from the aortic sac to the paired dorsal aortae, which fuse caudally into one descending aorta; the arteries appear in craniocaudal order during weeks 4 and 5 and are never all present at the same time.

· Truncus arteriosus gives the ascending aorta and the pulmonary trunk.

· Aortic sac: the right horn gives the brachiocephalic trunk and the left horn gives the arch as far as the left common carotid artery.

· First and second arch arteries: part of the maxillary artery, and the hyoid and stapedial arteries, with most of these vessels regressing.

· Third arch artery: the common carotid and the first part of the internal carotid, the rest of the internal carotid coming from the cranial dorsal aorta, and the external carotid sprouts from it.

· Fourth arch artery on the left: the arch between the left common carotid and the left subclavian arteries. On the right: the proximal right subclavian artery, with its distal part from the right dorsal aorta and the seventh intersegmental artery.

· Fifth arch artery: no derivative.

· Sixth arch arteries: the proximal right and left pulmonary arteries; the distal left becomes the ductus arteriosus and the distal right regresses.

· Left seventh intersegmental artery becomes the left subclavian artery, and the left dorsal aorta becomes the descending aorta beyond the left subclavian.

· Segments that disappear: the dorsal aorta between the third and fourth arches on both sides, and the right dorsal aorta below the right seventh intersegmental artery.

· The vagus supplies the fourth arch through the superior laryngeal nerve, which serves the cricothyroid muscle, and the sixth arch through the recurrent laryngeal nerve, which serves all other intrinsic laryngeal muscles.

· Each recurrent nerve first hooks under its sixth arch artery and is dragged caudally as the heart descends. On the right the distal sixth and the fifth arch arteries regress, so the nerve slides up to the right subclavian artery, a fourth arch derivative, at the root of the neck. On the left the distal sixth arch persists as the ductus, so the nerve stays under the aortic arch just behind the ligamentum arteriosum in the superior mediastinum. Both ascend in the tracheo-oesophageal groove.

Distinctions that matter

· The asymmetry of the two nerves comes from the asymmetry of the arch arteries, so the nerve course is a record of which vessel persisted.

· The internal carotid has two sources, the third arch artery proximally and the cranial dorsal aorta distally, so it is not a pure third arch derivative.

Common misconceptions

· Looking for a fifth arch derivative. There is none.

· Placing the right recurrent laryngeal nerve under the brachiocephalic trunk. It hooks under the right subclavian artery.

Where it is taught: Section C07, first paragraph, the table of origins of the great arteries and the paragraph on the laryngeal nerves of the CV-1 Cardiovascular Embryology notes.

NPLEX-P1-CARD-01-01-C07-BD2

Brain dump 2 · integration and application: Four remodelling errors and a hoarse voice

A competent response contains

· Aberrant right subclavian artery: the right fourth arch artery and the cranial right dorsal aorta regress while the caudal right dorsal aorta persists, so the right subclavian becomes the last branch of the arch and crosses behind the oesophagus, which can cause dysphagia.

· Double aortic arch: the right dorsal aorta persists, leaving a vascular ring that can compress the trachea and the oesophagus.

· Interrupted aortic arch between the left common carotid and the left subclavian: abnormal regression of the left fourth arch artery, so the lower body depends on a patent ductus arteriosus; because neural crest supports the fourth arch artery, this type has the strongest link with the 22q11.2 deletion.

· Nonrecurrent laryngeal nerve: with an aberrant right subclavian artery there is no right fourth arch vessel to hold the nerve down, so the right laryngeal nerve does not recur.

· The recurrent nerve supplies the posterior cricoarytenoid, the only abductor of the vocal fold, so injury leaves the fold near the midline and the voice is hoarse.

· Left nerve course in the chest: near the aortic arch, the left pulmonary artery, the left main bronchus and the hilar nodes, so hoarseness can be caused by an aortic arch aneurysm, a left hilar or mediastinal tumour or nodes, a dilated left atrium or pulmonary artery, which is cardiovocal or Ortner syndrome, or ligation of a patent ductus arteriosus.

· The right nerve is at risk at the root of the neck from an apical lung tumour or a subclavian aneurysm.

· Thyroid surgery puts both nerves at risk.

Distinctions that matter

· An aberrant right subclavian artery and a nonrecurrent right laryngeal nerve are two consequences of the same remodelling error, so they travel together.

· Only the left nerve has a long thoracic course, which is why thoracic and cardiac disease causes left sided vocal fold paralysis.

Common misconceptions

· Expecting a vascular ring to come from a persisting fourth arch artery. It comes from persistence of the right dorsal aorta.

· Attributing hoarseness in mitral disease to the voice box itself. A dilated left atrium and pulmonary artery compress the left recurrent nerve against the arch and the ligamentum arteriosum.

Where it is taught: Section C07, final paragraph on abnormal remodelling, the paragraph on nerve vulnerability and the stop and predict box of the CV-1 Cardiovascular Embryology notes.

NPLEX-P1-CARD-01-01-C02-Q01

Board-style question: An atrial opening that seals too early

Correct answer: C. Most left sided filling normally arrives across that opening, and chambers grow in proportion to the volume that passes through them

Why it is right: In the fetus pulmonary venous return is small, so the foramen ovale provides most left sided filling; when it seals early the left atrium and left ventricle receive little flow and grow poorly, and the right heart takes the volume and enlarges.

A is wrong: Oxygenated blood is not diverted into the pulmonary arteries, because high fetal pulmonary vascular resistance keeps pulmonary flow to a small fraction of combined output and the right ventricle still sends most of its output across the ductus arteriosus.

B is wrong: Left atrial pressure does not rise above right in this situation; with the opening sealed the left atrium is underfilled, and a small thick walled ventricle here reflects lost filling rather than an increased ejection load.

D is wrong: The endocardial cushions fuse and divide the atrioventricular canal in week 5, long before 24 weeks, and the mitral orifice is already formed, so cushion fusion is not what fails here.

A competent response contains

· Most left atrial and left ventricular filling in the fetus arrives across the foramen ovale, because pulmonary venous return is small while pulmonary vascular resistance is high.

· Embryonic and fetal chambers grow in proportion to the volume of blood that passes through them, so losing that filling leaves a small left atrium and a small left ventricle while the right heart enlarges.

Distinctions that matter

· Growth of a fetal chamber follows the flow through it, not the pressure it generates.

Common misconceptions

· Expecting an obstructed inflow to enlarge a chamber. Reduced filling leaves it underdeveloped.

Where it is taught: Section C02, stop and predict box, with the fetal circulation in Section C05 of the CV-1 Cardiovascular Embryology notes.

NPLEX-P1-CARD-01-01-C07-Q02

Board-style question: Hoarseness with a dilated aortic arch

Correct answer: A. The distal part of the left sixth pharyngeal arch artery

Why it is right: The distal left sixth pharyngeal arch artery persists as the ductus arteriosus and becomes the ligamentum arteriosum, which keeps the left recurrent laryngeal nerve in the chest beneath the aortic arch where an arch aneurysm can compress it.

B is wrong: The left fourth pharyngeal arch artery becomes the aortic arch between the left common carotid and left subclavian arteries; the nerve passes under the arch but it is held there by the ligament behind it rather than by the arch itself, and on the right it is the fourth arch derivative that holds the nerve at the neck.

C is wrong: The left seventh intersegmental artery becomes the left subclavian artery, which is the vessel the right nerve hooks under, not the structure that anchors the left nerve in the chest.

D is wrong: The left horn of the aortic sac becomes the arch up to the left common carotid artery and has no relationship that traps the recurrent laryngeal nerve.

A competent response contains

· The left recurrent laryngeal nerve first hooks under the left sixth arch artery; the distal part of that artery persists as the ductus arteriosus and leaves the ligamentum arteriosum behind it in the adult.

· That remnant holds the nerve in the superior mediastinum under the aortic arch, so an arch aneurysm compresses it and the left vocal fold is paralysed near the midline because the posterior cricoarytenoid, the only abductor, is denervated.

Distinctions that matter

· The right nerve has no equivalent thoracic anchor, because the distal right sixth arch artery regresses and the nerve rises to the right subclavian artery.

Common misconceptions

· Assuming both recurrent nerves loop around the same structure. The vessel that persisted on each side decides where the nerve ends up.

Where it is taught: Section C07, paragraph on the laryngeal nerves and the paragraph on nerve vulnerability of the CV-1 Cardiovascular Embryology notes.